

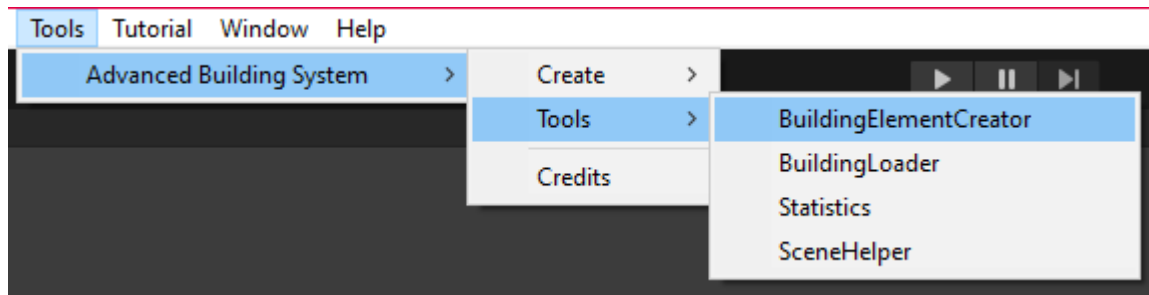
Tools

Building Element Creator.....	2
How it is work?.....	2
Properties.....	3
Instantiate.....	4
Building Loader.....	5
ABS Statistics.....	6
Scene Helper.....	7
Objects.....	7
Analysis.....	8

Building Element Creator

The `ABS_BuildingElementCreator` is a specific tool created for the Advanced Building System. The goal of the creator is to provide a tool that can be used to create `ABS_BuildingElement`. It is made to make the developers' workflow with creating the elements faster.

The creator can be found at:



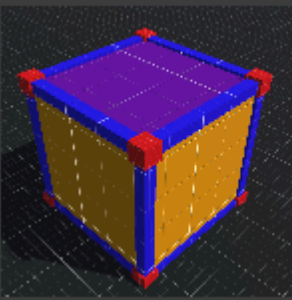
How it is work?

The creator works in that way a `GameObject` has to be provided with the algorithm settings that should be used for that element and the creator will create a building element.

The creation process is the following:

1. Create the building element's `GameObject`
2. Instantiate the given `GameObject` as a child object to the new `GameObject`
3. Add a `BoxCollider` and add the "Layer Of Player" layer to the collider's include list.
 - a. The layer is used from the given setting's layer collection.
4. Add the `ABS_BuildingElement` component
 - a. Set the given settings
 - b. Generate new `PrefabGuid`
 - c. Refresh the `BuildingElement`'s linkage which means that a `ABS_BuildingElementLink` component will added to every children `GameObject` what has a collider component
5. Copy the Copy Target's properties
6. Set the building element layer
 - a. The new building element's `GameObject`'s and every children's layer will be set to the "Layer Of BuildingElement" layer of the settings' Layer Collection

Properties



BuildingElement Creator

Use Object Name ☐

New BuildingElement Name

Algorithm Settings

 Missing MetaData

GameObject

 Missing GameObject

Copy Target

Use Object Name + New BuildingElement Name

If the “Use Object Name” boolean is true then the new building element’s name will be the provided “New BuildingElement Name” string. But if it is false then the given GameObject’s name will be used with a “_BuildingElement” postfix like:

```
m_BuildingElementName = $"{m_GameObject.name}_BuildingElement";
```

Algorithm Settings

An algorithm settings ScriptableObject should be provided that will be used during the creation process and also the BuildingElement will get it as a property.

GameObject

The given GameObject will be instantiated as a child for the new building element.

Copy Target

It can be null but if it is provided the given building element’s properties will be copied to the new building element.

The following properties will be copied:

- FinalElement
- PreBuilt
- Foundation
- Indestructible
- AreaType
- SnapToPreBuiltFinalElement
- ShouldAllowedByArea

- ShouldOverride
- DragBuildingEnabled
- DragBuildingBehaviour
- EnabledDragBuildingX
- EnabledDragBuildingZ
- HighlightCollection
- CanNotBeAttachTarget
- ShouldAttached
- StableElement
- AdvancedGridSnapPointRuleSet

Also the BoxCollider's size will be set to the Copy Target's Dimension Vector3

Instantiate

If every property has provided the following to options are available:

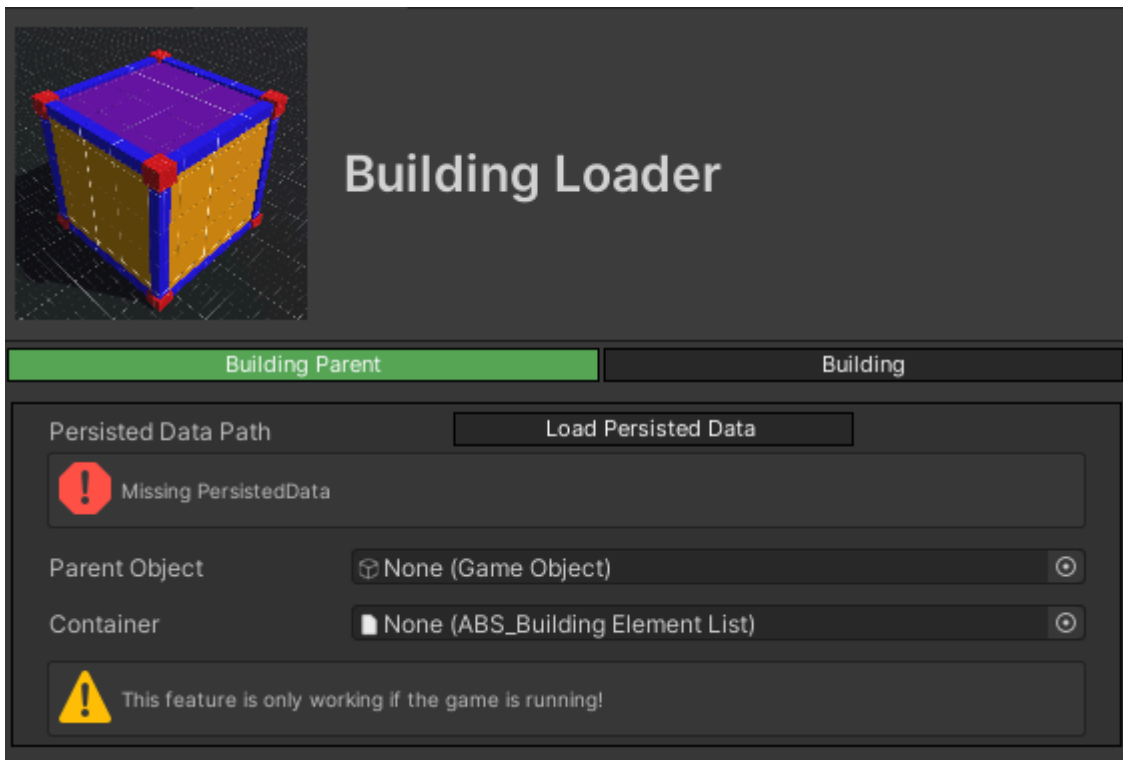
- Instantiate GameObject
With this option the ABS_BuildingElement will be instantiated into the scene.
- Create Prefab
With this option a prefab should be created to the selected path about the ABS_BuildingElement.
Note that for this option the tool should instantiate the ABS_BuildingElement just like the other option but after the creation of the prefab the GameObject is removed immediately.

Building Loader

The BuildingLoader is a complementary tool for the persistence functionality of the BuildingManager. If you have a loading file for a Building/BuildingParent with this tool you can load it into your scene. Find it under the Tools > AdvancedBuildingSystem > Tools.

The tool works in that way you have to select a PersistedData file and a BuildingElementList (which contains every BuildingElement from the Buildings) should be provided and just press the Load button. It will load the persisted Building or BuildingParent into your scene. Also you can give a parent object from the Hierarchy .

Note that this tool can be used only when the game is running. If you do not want to run the scene this tool works totally fine with an empty scene.

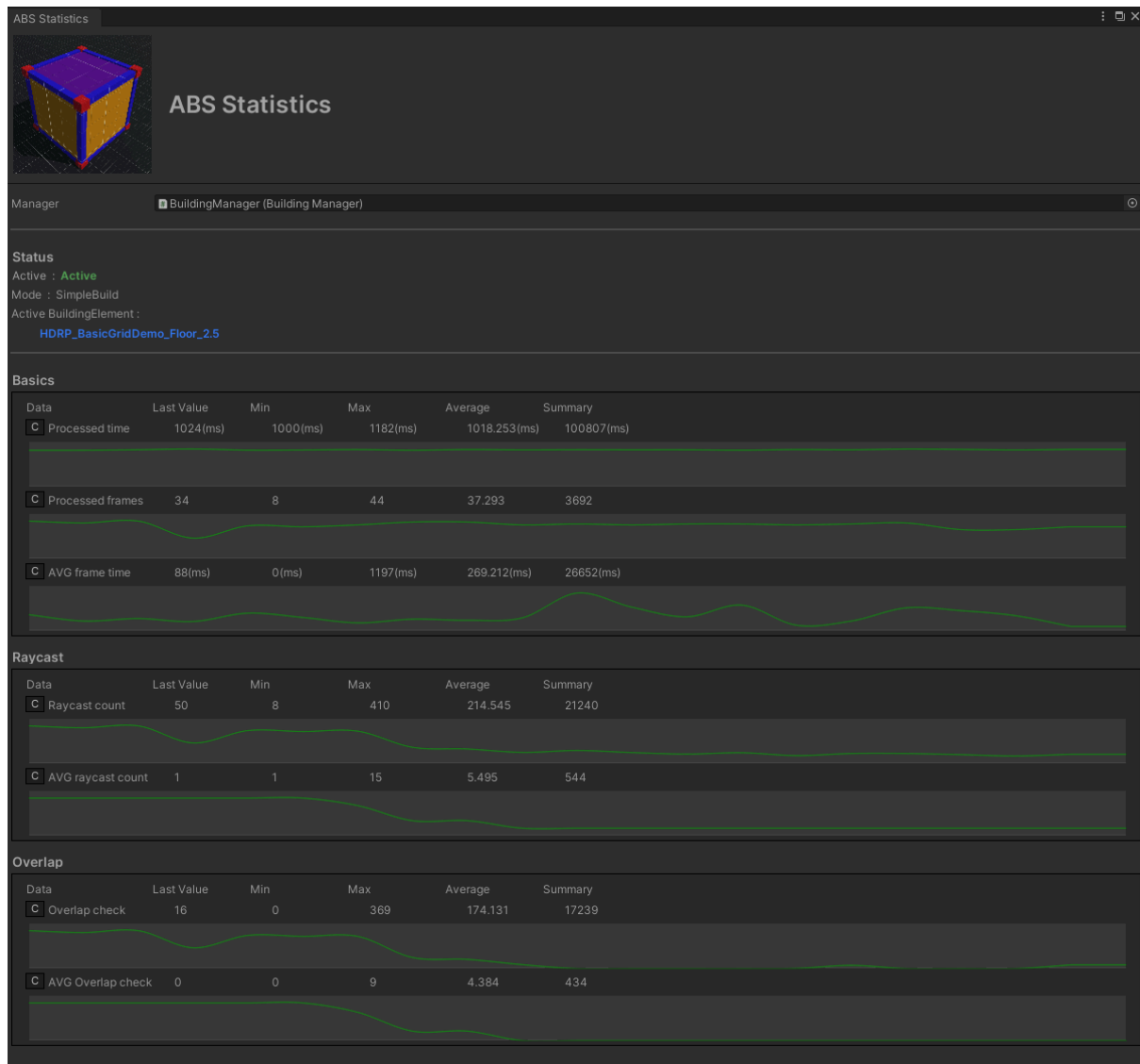


ABS Statistics

The ABS Statistics is based on the IBuildingManagerStatisticsTracker. This tool gives a visual representation of the data of the Statistics feature.

This tool amplify the data of the Statistics with:

- Minimum Value
- Maximum Value
- Average Value
- Summary



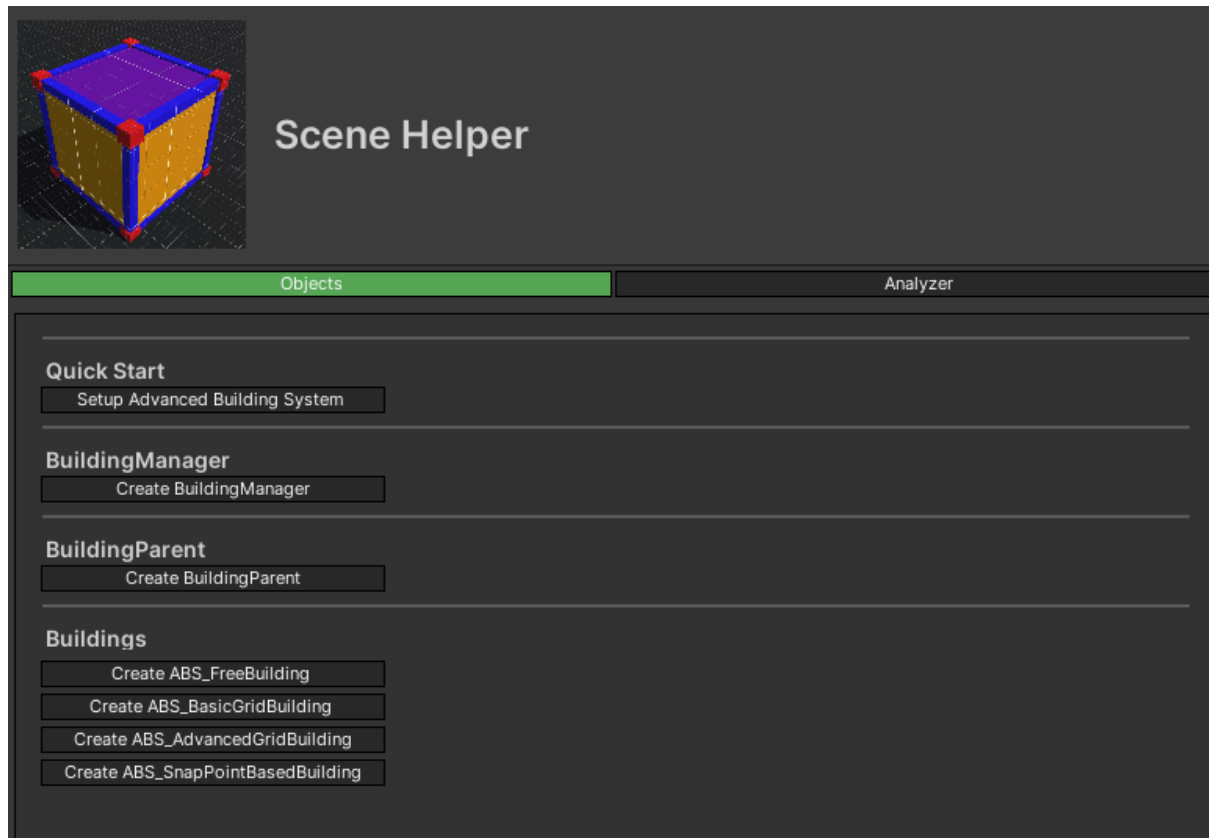
Scene Helper

The purpose with the Scene Helper is helping the scene building process for the developers.

Objects

This tab of the Scene helper is for creating ABS GameObject into the scene like buildings and BuildingManager etc.

Every property for the created ABS objects are set if possible.



Also this tab has a Quick Start option that setup a whole AdvancedBuildingSystem into your scene. It creates the BuildingManager, BuildingParent with the necessary global buildings and sets every property possible. The SceneHelper Analyzer should only find one issue which is the BuildingElementList that should be provided by the developer. This feature is for speeding up the process of setting up the scene.

Analysis

It analyzes the active scene and identifies errors. If the tool can fix an error, it provides hotfix options.

The GameObject names are links on the UI that open the objects in the Inspector window.

